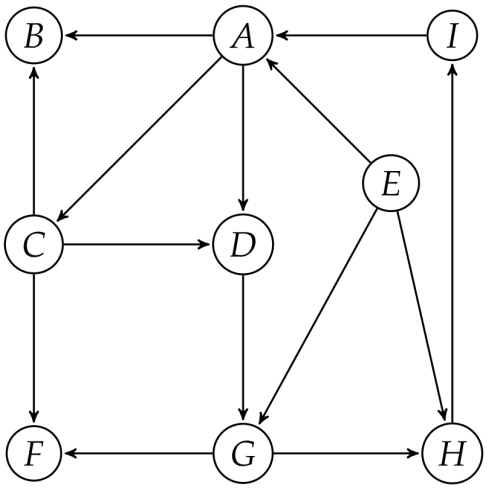
CSE221 Quiz

Total Marks: 10

Problem #1 Run depth-first search on the graph below starting at node A. Whenever you have a choice of vertices, pick the one that appears earlier alphabetically. 5



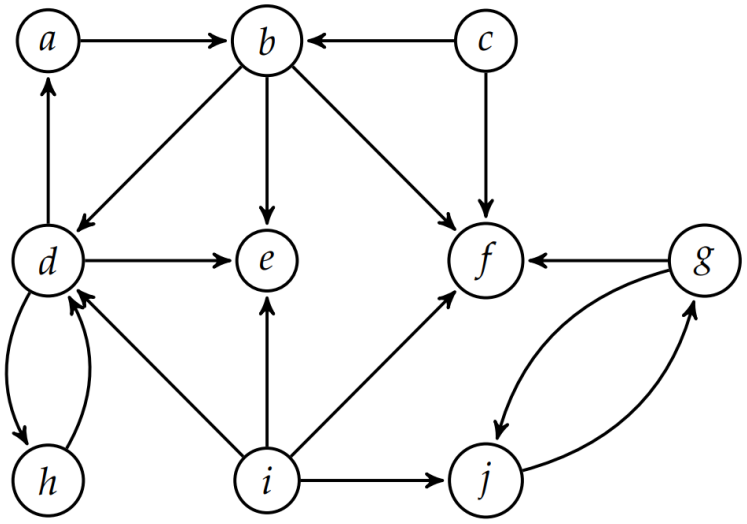
a) Now write down the start time and finish time values for each vertex in the following table. 2

Vertex	Start time	Finish time
A		
B		
C		
D		
E		
F		
G		
H		
I		

b) for each edge indicate using ✓ marks if it is a tree edge, a forward edge, a back edge, or a cross. 3

Edge	tree	forward	back	cross
$A \to B$				
$A \to C$				
$A \to D$				
$C \to B$				
$C \to D$				
$C \to F$				
$D \to G$				
$G \to F$				
$G \to H$				
$H \to I$				
$E \to A$				
$E \to G$				
$E \to H$				
$I \to A$				

Problem #2 Consider the following directed unweighted graph. You will compute strongly connected components in the following graph. 5



a) Run DFS and sort the vertices according to finish time. Write down the sorted list. 2

b) Now reverse the edges and retrieve its strongly connected components one at a time. At each step, write down what vertex you are calling DFS on, and the vertices in the connected component retrieved. 3

Starting vertex	Vertices in the connected component

* Please note that there might be more rows in the table above than necessary

Submission link: https://forms.gle/X5i14p6FP9JnSkki9 (Deadline: 11:59PM)